

MAURICE BISHOP INTL, GrenadaWMO: **789580**Lat: **12.004N**Lon: **61.786W**Elev: **23**StdP: **14.68**Time Zone: **-4.00 (W04)**Period: **94-19**WBAN: **00409****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.1	73.9	67.1	99.8	78.0	67.9	102.7	78.0	24.8	82.6	23.3	82.4	5.0	60	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.6	88.5	80.4	88.1	80.3	87.8	80.2	82.1	86.5	81.4	86.1	80.9	85.8	13.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	159.8	86.2	80.2	156.7	85.7	79.2	151.6	84.8	45.9	87.3	45.2	86.6	44.7	87.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	21.5	20.4	DB	69.7	91.5	3.2	2.9	67.4	93.6	65.6	95.2	63.8	96.8	61.5	98.9	(4)
(5)			WB	67.2	83.8	2.7	1.2	65.2	84.7	63.7	85.4	62.1	86.1	60.2	86.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	79.8	79.8	80.5	81.8	82.6	82.3	82.2	82.5	83.3	82.7	81.9	80.8
	DBStd	1.79	1.32	1.33	1.44	1.39	1.24	1.45	1.29	1.39	1.56	1.56	1.46	1.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11567	923	835	946	953	1011	970	997	1009	999	1013	956	955
	CDD65	6092	458	415	481	503	546	520	532	544	549	548	506	490
	CDH74	64383	3960	3589	4495	5264	6154	5793	5897	6273	6551	6271	5407	4729
CDH80	18394	791	710	997	1392	1850	1677	1717	2063	2419	2104	1563	1111	
Wind	WSAvg	11.9	11.5	13.1	13.3	14.2	14.4	14.2	11.8	10.1	10.2	10.0	10.0	10.8
Precipitation	PrecAvg	63.1	4.3	2.3	1.9	2.4	3.4	6.5	6.7	7.4	6.4	7.2	9.1	5.6
	PrecMax	78.8	13.9	6.1	4.3	10.4	9.3	11.1	10.7	10.0	12.9	14.4	19.6	17.8
	PrecMin	45.7	1.5	0.4	0.3	0.6	1.0	3.1	3.5	4.5	2.8	3.0	3.4	1.6
	PrecStd	6.7	2.5	1.3	1.1	2.2	2.4	2.2	1.6	1.6	2.3	2.7	3.6	3.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	86.3	86.4	88.0	88.2	88.1	88.2	89.4	89.9	89.6	88.2	87.6
		MCWB	76.7	76.4	77.2	79.3	79.7	79.3	79.6	80.5	81.1	80.8	80.2	78.9
	2%	DB	85.5	85.2	86.0	86.4	87.6	87.5	87.6	88.1	88.9	88.2	87.8	86.3
		MCWB	76.5	76.2	77.1	78.3	79.5	79.3	79.6	80.5	80.8	80.6	80.1	78.3
	5%	DB	84.5	84.4	84.9	86.1	86.4	86.3	86.4	87.6	88.1	87.8	86.4	85.9
		MCWB	76.3	75.9	76.5	78.0	78.8	78.9	79.2	80.3	80.6	80.6	79.5	78.2
	10%	DB	84.1	84.0	84.3	85.6	86.0	85.9	86.0	86.4	87.7	86.9	86.1	84.6
		MCWB	76.2	75.7	76.2	77.7	78.7	78.8	79.1	79.7	80.5	80.2	79.3	77.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.2	79.4	79.9	81.0	81.9	81.7	81.6	82.5	82.8	82.6	82.0	80.7
		MCDB	83.8	83.7	84.6	85.8	85.9	85.6	85.6	86.7	87.5	87.1	86.2	85.3
	2%	WB	78.3	78.2	78.7	80.0	80.8	80.6	80.8	81.4	81.9	81.7	81.0	79.8
		MCDB	83.3	83.3	84.1	85.1	85.2	85.1	85.3	86.0	86.9	86.5	85.6	84.6
	5%	WB	77.6	77.3	78.0	79.2	80.1	80.0	80.1	80.8	81.1	81.0	80.5	79.1
		MCDB	82.8	82.8	83.4	84.4	84.6	84.5	84.7	85.7	86.4	85.9	85.2	84.0
	10%	WB	76.8	76.4	77.3	78.4	79.5	79.5	79.8	80.3	80.7	80.5	79.9	78.3
		MCDB	82.2	82.0	82.8	83.8	84.2	84.0	84.3	85.2	86.1	85.5	84.5	83.2
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.0	8.7	7.9	7.4	7.4	8.1	8.5	8.6	8.8	8.9	9.1
		MCDBR	9.6	9.3	9.0	8.2	7.7	7.8	8.5	9.2	9.1	9.2	9.3	9.6
		MCWBR	4.9	4.8	4.6	4.3	4.1	4.1	4.4	4.9	4.8	4.9	4.9	5.1
	5% WB	MCDBR	8.9	8.7	8.2	7.6	7.4	7.7	8.2	8.6	8.7	8.8	8.9	9.0
		MCWBR	5.1	5.2	4.8	4.5	4.4	4.4	4.6	5.1	4.9	5.1	5.0	5.2
Clear-Sky Solar Irradiance	taub	0.379	0.398	0.399	0.441	0.470	0.531	0.500	0.442	0.420	0.389	0.381	0.388	
	taud	2.585	2.506	2.516	2.404	2.309	2.108	2.220	2.447	2.502	2.621	2.633	2.584	
	Ebn at Noon	294	291	292	276	264	245	254	272	281	290	291	288	
	Edh at Noon	31	35	35	39	42	50	45	37	35	30	30	30	
All-Sky Solar Radiation	RadAvg	1720	1929	2056	2103	2041	1892	1945	2000	1980	1800	1643	1616	
	RadStd	66	63	90	81	123	97	92	70	76	83	114	79	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.36	+0.59	+0.74	+0.25	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page